

# Slide 3 Reading Notes

Analytical approach | Meds Consulting LLC

## Opening

This slide explains the analytical approach we will use after the initial data review. The goal is to move from data quality checks into an analysis that can support practical decisions for NHN.

### 1. Why this approach fits the NHN problem

Readmissions are not caused by one single factor. They are shaped by clinical risk, care-transition activity, discharge setting, utilization history, and financial exposure. Because the problem has multiple layers, we need an approach that can explain risk, prioritize patient groups, and validate where the strongest opportunities are.

The approach has three parts: regression analysis, patient segmentation, and market or baseline analysis. Together, these methods help us connect statistical evidence with operational action.

### 2. Regression analysis

First, regression analysis helps us quantify which factors are associated with readmission. We can model readmission likelihood using patient characteristics, utilization history, discharge information, and care coordination variables.

This is useful because it controls for overlap between variables. For example, older patients may also have more chronic conditions or prior admissions. Regression helps us understand which factors still matter after accounting for those relationships.

The output should be explainable, not just technical. We want risk indicators that leadership and clinical teams can understand, such as odds ratios, feature importance, or predicted readmission risk.

### 3. Patient segmentation

Second, patient segmentation helps translate the analysis into action. Instead of treating all patients as one average group, we can create meaningful segments by age, chronic disease burden, prior admissions, discharge setting, and follow-up status.

Then we can compare readmission rates and cost exposure across those groups. This helps NHN identify which patient cohorts should receive additional care coordination first.

The practical output is a high-risk cohort list and an intervention priority list. This is important because NHN needs to decide where limited resources can have the largest effect.

### 4. Market and baseline analysis

Third, market and baseline analysis puts the results in context. In this project, that means comparing NHN patterns across service lines, discharge types, and care-transition activity rather than only looking at one overall readmission rate.

This helps separate broad readmission pressure from segment-specific performance gaps. It also gives us a basis for setting realistic improvement targets and financial scenarios.

The output could include a benchmark table, a trend summary, and savings scenarios that show what happens if NHN reduces avoidable readmissions in targeted groups.

## **5. Decision standard**

The decision standard is that recommendations should be statistically supported, clinically explainable, operationally feasible, and financially material.

In other words, we are not only looking for variables that are mathematically significant. We are looking for actions NHN can actually implement to reduce avoidable readmissions and improve patient outcomes.